

Sales & E-Commerce SQL Practice Dataset

MySQL-ready relational dataset + Basic → Advanced SQL interview/project questions

Dataset design: 10 stores, 5 regions, 8 categories, 20 suppliers, 60 products, 30 employees, 300 customers, 1,000 orders and 2,500+ order-detail rows. Keys are internally consistent and suitable for JOIN, CTE, window-function and optimization practice.

1. Database Schema

Table	Rows	Primary Key	Important Foreign Keys
regions	5	region_id	—
stores	10	store_id	region_id → regions
categories	8	category_id	—
suppliers	20	supplier_id	—
products	60	product_id	category_id → categories; supplier_id → suppliers
employees	30	employee_id	region_id → regions
customers	300	customer_id	—
orders	1,000	order_id	customer_id; store_id; employee_id
order_details	2,500+	order_detail_id	order_id → orders; product_id → products

2. Column Definitions

Table	Columns
regions	region_id, region_name
stores	store_id, store_name, region_id
categories	category_id, category_name
suppliers	supplier_id, supplier_name, supplier_city
products	product_id, product_name, category_id, supplier_id, unit_price, unit_cost
employees	employee_id, employee_name, region_id, job_title
customers	customer_id, customer_name, city, region_name, segment, signup_date
orders	order_id, order_date, customer_id, store_id, employee_id, status, payment_method
order_details	order_detail_id, order_id, product_id, quantity, unit_price, discount_pct

3. Sample Dataset

customers — first 25 rows

customer_id	customer_name	city	region	segment	signup_date
1	Customer 001	Hyderabad	East	Consumer	2025-06-10
2	Customer 002	Bengaluru	South	Corporate	2025-11-16
3	Customer 003	Chennai	West	Consumer	2025-07-06
4	Customer 004	Mumbai	South	Small Business	2025-08-07
5	Customer 005	Pune	Central	Corporate	2024-09-05
6	Customer 006	Delhi	West	Corporate	2024-07-13
7	Customer 007	Kolkata	North	Consumer	2025-11-05
8	Customer 008	Jaipur	West	Corporate	2025-03-09
9	Customer 009	Ahmedabad	West	Corporate	2024-02-25
10	Customer 010	Indore	North	Consumer	2025-02-16
11	Customer 011	Hyderabad	East	Consumer	2024-09-11
12	Customer 012	Bengaluru	South	Consumer	2025-07-03
13	Customer 013	Chennai	West	Consumer	2025-03-08
14	Customer 014	Mumbai	South	Corporate	2025-04-18
15	Customer 015	Pune	South	Consumer	2025-03-29
16	Customer 016	Delhi	Central	Consumer	2024-02-21
17	Customer 017	Kolkata	Central	Consumer	2024-04-05
18	Customer 018	Jaipur	South	Consumer	2025-02-20
19	Customer 019	Ahmedabad	West	Corporate	2024-08-06
20	Customer 020	Indore	West	Consumer	2024-06-17
21	Customer 021	Hyderabad	West	Consumer	2025-02-03
22	Customer 022	Bengaluru	East	Corporate	2024-10-19
23	Customer 023	Chennai	West	Small Business	2025-07-23
24	Customer 024	Mumbai	West	Consumer	2024-07-13
25	Customer 025	Pune	East	Consumer	2024-02-29

products — first 25 rows

product_id	product_name	category_id	supplier_id	unit_price	unit_cost
1	Wireless Mouse	2	1	962.02	441.81
2	Mechanical Keyboard	2	2	418.34	222.8
3	USB-C Hub	3	3	1106.82	775.19
4	27in Monitor	2	4	1339.13	645.68
5	Laptop Stand	2	5	637.51	293.91
6	Bluetooth Speaker	8	6	334.21	212.89
7	Noise Cancelling Headphones	8	7	47.59	24.92
8	Webcam	2	8	977.63	637.05
9	Power Bank	3	9	336.9	225.06
10	Smartphone Case	3	10	1215.67	549.97
11	Fast Charger	3	11	1210.28	857.26

product_id	product_name	category_id	supplier_id	unit_price	unit_cost
12	External SSD	2	12	515.65	261.71
13	WiFi Router	2	13	1436.16	825.13
14	Office Chair	6	14	146.38	71.11
15	Desk Lamp	5	15	1272.46	856.85
16	Gaming Mouse	7	16	1212.24	872.81
17	Gaming Keyboard	7	17	808.05	654.56
18	HDMI Cable	3	18	572.77	374.74
19	Air Purifier	4	19	1245.47	845.49
20	Coffee Maker	4	20	1293.67	858.51
21	Notebook Pack	5	1	1059.22	494.61
22	Desk Organizer	5	2	348.02	193.87
23	Bookshelf	6	3	127.05	68.12
24	Standing Desk	6	4	158.69	87.73
25	Game Controller	7	5	956.44	559.51

orders — first 30 rows

order_id	order_date	customer_id	store_id	employee_id	status	payment
1	2025-01-30	78	8	2	Completed	Cash
2	2024-05-27	210	3	3	Completed	Debit Card
3	2025-05-14	278	1	20	Completed	UPI
4	2024-11-02	15	1	11	Completed	Debit Card
5	2024-06-28	41	10	28	Completed	Cash
6	2025-10-15	131	8	9	Returned	Credit Card
7	2024-09-13	234	5	7	Completed	Net Banking
8	2025-01-02	295	5	23	Completed	Credit Card
9	2025-10-13	98	10	9	Returned	Wallet
10	2025-08-16	187	3	3	Completed	Debit Card
11	2024-10-29	174	2	15	Completed	UPI
12	2025-02-08	8	5	18	Completed	Net Banking
13	2024-01-26	288	6	30	Cancelled	UPI
14	2024-11-05	210	2	5	Completed	Credit Card
15	2025-11-17	277	7	19	Returned	Wallet
16	2024-10-13	17	6	7	Completed	Net Banking
17	2025-10-22	184	1	13	Completed	UPI
18	2025-11-09	109	10	1	Completed	Debit Card
19	2024-07-31	111	4	11	Completed	Cash
20	2025-07-07	129	3	4	Returned	Credit Card
21	2024-06-27	136	1	5	Returned	Net Banking
22	2025-04-04	186	9	19	Completed	Net Banking
23	2024-02-01	32	8	28	Completed	Net Banking
24	2024-03-23	165	10	5	Completed	UPI

order_id	order_date	customer_id	store_id	employee_id	status	payment
25	2024-10-28	233	9	20	Completed	Credit Card
26	2024-08-08	221	8	29	Completed	Net Banking
27	2025-11-12	131	6	5	Returned	Net Banking
28	2025-03-18	50	6	26	Completed	Cash
29	2024-12-03	63	7	12	Returned	Net Banking
30	2024-12-26	54	10	17	Completed	UPI

order_details — first 40 rows

detail_id	order_id	product_id	qty	unit_price	discount
1	1	56	4	1314.77	0
2	1	7	5	47.59	0.1
3	1	55	1	684.96	0.2
4	2	40	1	403.2	0.05
5	2	45	5	143.64	0.1
6	2	26	3	560.31	0.2
7	3	15	4	1272.46	0
8	3	48	1	102.78	0.1
9	3	6	2	334.21	0.2
10	4	24	2	158.69	0.15
11	4	28	4	974.87	0.15
12	4	10	2	1215.67	0
13	5	32	5	838.97	0
14	5	59	2	453.28	0.1
15	6	58	2	1369.64	0
16	6	19	4	1245.47	0.05
17	6	44	5	1496.33	0.05
18	6	35	4	478.66	0.2
19	7	16	4	1212.24	0.15
20	8	18	1	572.77	0.15
21	8	1	4	962.02	0.05
22	8	37	2	1315.54	0.15
23	8	56	3	1314.77	0
24	9	41	1	845.56	0.05
25	9	7	4	47.59	0
26	10	12	5	515.65	0.05
27	10	13	5	1436.16	0.15
28	10	9	3	336.9	0
29	10	51	3	25.13	0.1
30	11	56	4	1314.77	0.15
31	11	44	3	1496.33	0
32	12	44	5	1496.33	0.1
33	12	48	3	102.78	0
34	12	17	2	808.05	0.1
35	13	41	4	845.56	0.2
36	14	32	2	838.97	0.15
37	14	8	2	977.63	0.1
38	14	7	4	47.59	0.05
39	15	57	1	754.87	0.1
40	15	27	5	406.33	0.1

4. Derived Metrics

Line Revenue = quantity × unit_price × (1 – discount_pct)

Line Cost = quantity × products.unit_cost

Line Profit = Line Revenue – Line Cost

Profit Margin % = Line Profit / Line Revenue × 100

For cancelled/returned orders, decide whether to exclude them and state your assumption. This is intentional: it gives you realistic business-rule practice.

5. SQL Practice Questions — 120 Problems

Recommended progression: solve Levels 1–2 without CTEs or window functions, Levels 3–4 with CTEs/window functions, then use EXPLAIN and indexes for Level 5. For Level 6, write production-style SQL and document assumptions.

Level 1 — Basic SQL

1. Display all customers.
2. Display only customer name, city and segment.
3. Find all products with unit_price greater than 500.
4. Find products priced between 100 and 500.
5. Find customers from Hyderabad.
6. Find customers belonging to the Corporate segment.
7. List all distinct payment methods.
8. List all distinct order statuses.
9. Find orders placed during 2025.
10. Sort products from highest to lowest unit_price.
11. Display the 10 cheapest products.
12. Find products whose name contains 'Mouse'.
13. Find customers whose name ends with '5'.
14. Count the total number of customers.
15. Count the total number of products.
16. Find the minimum, maximum and average product price.
17. Find the total quantity sold.
18. Calculate total gross sales before discount.
19. Find the number of orders per status.
20. Find the number of customers per segment.
21. Find the average product price by category_id.
22. Find categories having more than 5 products.
23. Find payment methods with more than 100 orders.
24. Find the total quantity sold for each product.
25. Find the total revenue for each product using order_details.

Level 2 — Joins & Business Analysis

26. Join products with categories and display product and category names.
27. Join products with suppliers and display supplier names.
28. Join stores with regions.
29. Join employees with regions.
30. Join orders with customers.
31. Join orders with stores and regions.
32. Join order_details with products.
33. Calculate revenue by category.
34. Calculate revenue by region.
35. Calculate revenue by store.
36. Calculate revenue by salesperson.
37. Find the top 10 customers by revenue.
38. Find the top 10 products by revenue.
39. Find the top 5 stores by revenue.

40. Find the top 5 sales employees by revenue.
41. Find customers who have placed at least one order.
42. Find customers who have never placed an order.
43. Find products that have never been ordered.
44. Find suppliers whose products generated more than 100,000 in revenue.
45. Find the number of orders handled by each employee.
46. Find average order value by store.
47. Find average order value by customer segment.
48. Compare completed vs cancelled order counts.
49. Calculate revenue by payment method.
50. Find the best-selling product in each category.

Level 3 — Subqueries & CTEs

51. Find products priced above the overall average product price.
52. Find customers whose total revenue is above average customer revenue.
53. Find employees whose revenue is above average employee revenue.
54. Find stores whose revenue is above average store revenue.
55. Find the second-highest priced product.
56. Find the third-highest revenue-generating customer.
57. Find the category with the highest average product price.
58. Find products whose unit_price is greater than their category average.
59. Find customers who ordered every month in 2025.
60. Find customers with more than 3 completed orders.
61. Find products purchased by more than 50 distinct customers.
62. Using a CTE, calculate monthly revenue.
63. Using a CTE, calculate monthly revenue and month-over-month change.
64. Using a CTE, find the top 3 products by category.
65. Using a CTE, find the top 5 customers by region.
66. Using a CTE, calculate customer lifetime revenue.
67. Using multiple CTEs, calculate revenue, cost and profit.
68. Using a CTE, identify stores with below-average profit margin.
69. Using a CTE, identify repeat customers.
70. Using a CTE, calculate each category's contribution to total revenue.

Level 4 — Window Functions

71. Rank products by revenue using RANK().
72. Rank products by revenue within each category.
73. Rank employees by revenue within each region.
74. Use DENSE_RANK() to find the top 3 customers in each segment.
75. Use ROW_NUMBER() to return the first order of every customer.
76. Use ROW_NUMBER() to return the latest order of every customer.
77. Use LAG() to compare monthly revenue with the previous month.
78. Use LEAD() to find the next order date for each customer.
79. Calculate a running monthly revenue total.
80. Calculate a running revenue total for each region.
81. Calculate a 3-month moving average of revenue.

82. Calculate each product's percentage contribution to category revenue.
83. Calculate each region's percentage contribution to total revenue.
84. Find the highest-revenue product in every category using ROW_NUMBER().
85. Find the second-highest employee in every region.
86. Calculate the difference between an employee's revenue and regional average revenue.
87. Find customers whose current order value is greater than their previous order value.
88. Calculate cumulative customer revenue over time.
89. Calculate monthly order growth percentage.
90. Find the month with the largest revenue increase.

Level 5 — Advanced SQL & Optimization

91. Use EXPLAIN to inspect a query joining orders and order_details.
92. Create an index on orders(order_date) and compare execution plans before and after.
93. Create a composite index for a frequent customer/date filter and test it with EXPLAIN.
94. Compare a correlated subquery with a CTE for customer revenue.
95. Rewrite a query that uses SELECT * so it returns only required columns.
96. Identify which predicates should be applied before aggregation.
97. Write a query that filters completed orders before joining order_details.
98. Compare WHERE vs HAVING for filtering order-level rows.
99. Find duplicate customer records based on city, segment and similar names.
100. Find duplicate order-detail combinations for the same order and product.
101. Detect products where unit_cost is greater than or equal to unit_price.
102. Detect orders whose detail rows have inconsistent pricing.
103. Find orphaned order_details using a LEFT JOIN.
104. Find orphaned products/suppliers using referential checks.
105. Build a reusable view for completed sales.
106. Build a view for monthly revenue and profit.
107. Create a stored query/report that returns monthly KPIs.
108. Optimize a top-10-products revenue query.
109. Optimize a customer lifetime-value query.
110. Design indexes for the 5 most common analytical queries and explain why.

Level 6 — Interview / Case Study

111. A manager says revenue fell last month. Build a query to identify the largest regional and product contributors to the decline.
112. Identify customers whose order frequency is increasing over time.
113. Find the top 20% of customers and their share of total revenue.
114. Perform an 80/20 Pareto analysis of products by revenue.
115. Find products with high revenue but low profit margin.
116. Find products with high profit margin but low sales volume.
117. Find stores with above-average revenue and below-average margin.
118. Find employees with strong revenue but weak average order value.
119. Find the first purchase date and latest purchase date for every customer.
120. Calculate customer lifetime value using completed orders.
121. Calculate monthly active customers.
122. Calculate customer retention by signup month.
123. Calculate repeat-purchase rate.

124. Find customers who purchased in January but not February.
125. Find customers who purchased in both January and February.
126. Find customers who purchased in every quarter of 2025.
127. Find the longest gap between two orders for each customer.
128. Find the fastest-growing product category year over year.
129. Find the best and worst month by revenue and explain the calculation.
130. Create a final executive sales report query with revenue, orders, customers, AOV, profit and margin by month.

6. Recommended Project Workflow

Stage	What to do
1. Load	Create the 9 tables, PKs and FKs; insert the data.
2. Validate	Check row counts, NULLs, duplicates, orphan keys and invalid prices.
3. Basic analysis	Solve Levels 1–2 using SELECT, filters, aggregation and joins.
4. Advanced analysis	Solve Levels 3–4 with subqueries, CTEs and window functions.
5. Optimization	Run EXPLAIN, add targeted indexes and compare execution plans.
6. Portfolio	Create a SQL project with business questions, queries, findings and optimization notes.

Important: The PDF contains the schema, column definitions, representative sample rows and the full question bank. The underlying dataset is designed to be generated/imported as SQL data; the PDF is intentionally not filled with thousands of repetitive rows.